Probabilistic Machine Learning

Lecture 05: Bayesian Networks

Prof. Dr. Michael Granitzer

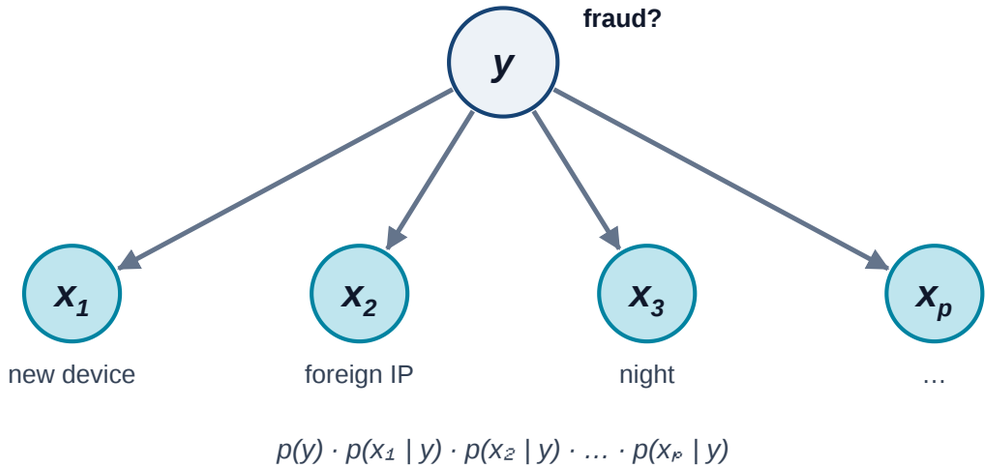


Speaker notes

The lecture's place in the arc: "graph the joint". Lecture 03 assumed every attribute apart; this lecture lets the modeller state exactly which independences hold, draws them, reads them off, and then asks what the drawing costs. Five modules: graph the joint (conditional independence counted, the wet-grass model built and drawn, explaining away); what an arrow claims (belief networks, parameter counts); reading independence (d-separation on the wet grass, Markov equivalence); plate notation (the two models of Lecture 07, drawn); and computing with the graph (a worked elimination, Markov networks as what inference runs on, "writable is not queryable" on a pixel lattice, learning as counting).

2026-09-26 (course plan N2): Module 1 moved here from Lecture 04. The shared Data Science units are replaced in PML by PML units with drawn figures; the shared units are unchanged. Markov networks are two slides inside Module 5; soft and unreliable evidence is in the appendix.

Where We Left Off



Lecture 03 made the joint affordable with one blanket assumption: **every** attribute independent of every other, given the class. It worked — and it counted correlated evidence twice.

What if we assumed only the independences we actually believe?

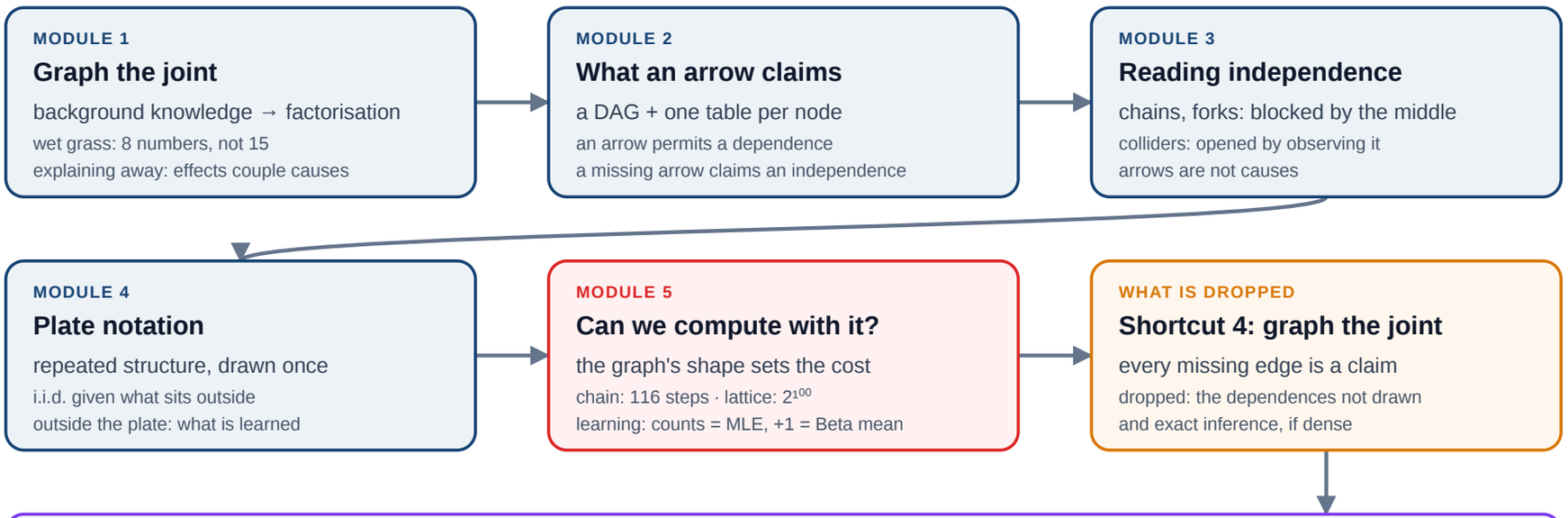
Shortcut 4 — graph the joint. Draw which variables depend on which; the graph *is* the factorisation. Dropped: every dependence **not drawn** — and, for a dense graph, affordable inference.

Speaker notes

Pick up the shortcut table of Lecture 01 again. "Assume it apart" was a blanket; "graph the joint" is the same move made selective: the modeller decides which independences to claim, and the graph states them. Say both costs now, so that the lecture can pay them off: a missing arrow is a claim that may be wrong (Modules 1–3), and a graph that can be written down may still be too expensive to condition on (Module 5, "writable is not queryable").

Argument Map

Lecture 05 · argument map



Course arc
ideal → shortcuts

L01	L02	L03	L04	L05	L06	L07
The ideal	The language	Peak · apart	Only p(y | x)	Graph the joint	Keep the loss	No shortcut

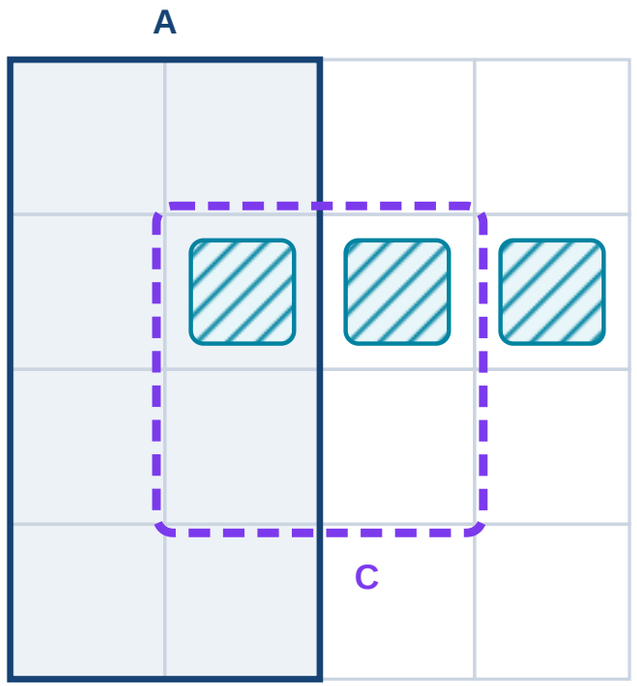
Speaker notes

The lecture on one page. Top row: the joint graphed, what its arrows claim, and how independence is read off. Middle row: the notation for repeated structure, the cost of querying, and the shortcut named with what it drops. The takeaway is the lecture in one sentence: writable is not queryable. Point at the strip: Lecture 05 is "graph the joint".

Graph the Joint

Guiding Question: Naive Bayes assumed every attribute independent of every other. What if we stated only the independences we believe — and what does conditioning then do to them?

Independent Given C — Counted



A 8 of 16 cells

B 3 of 16 cells

C 4 of 16 cells

Sixteen equally likely cells.

Dependent

$$P(A \cap B) = \frac{1}{16} \neq \frac{3}{32} = P(A)P(B)$$

Independent given C

$$P(A \cap B \mid C) = \frac{1}{4} = P(A \mid C)P(B \mid C)$$

Speaker notes

The definition from Lecture 03, checked by counting cells. Students can verify every number on the picture: inside the dashed block, half the cells are A, half are B, and exactly one quarter are both. Also directly: $P(A \mid B \cap C) = 1/2 = P(A \mid C)$.

Give the everyday version: ice-cream sales and drownings are dependent, but independent given the temperature — a common cause. Conditioning on it *removed* a dependence. The rest of the module shows the opposite case.

Why Is the Grass Wet?

Tracey sees her lawn is wet. **Rain — or did she leave the sprinkler on?** Then she sees that her neighbour Jack's lawn is wet too.

variable	meaning (1 = yes)
R	it rained
S	Tracey's sprinkler was on
J	Jack's grass is wet
T	Tracey's grass is wet

Before any arithmetic: does Jack's wet lawn make the sprinkler **more** or **less** likely?

Speaker notes

Barber's classic example. Collect predictions and keep them on the board; intuitions differ, which is the point. The answer comes two slides later, with numbers.

Background Knowledge as Independence

The joint needs $2^4 - 1 = 15$ numbers. The chain rule rewrites it without shrinking it:

$$p(T, J, R, S) = p(T \mid J, R, S) \cdot p(J \mid R, S) \cdot p(R \mid S) \cdot p(S)$$

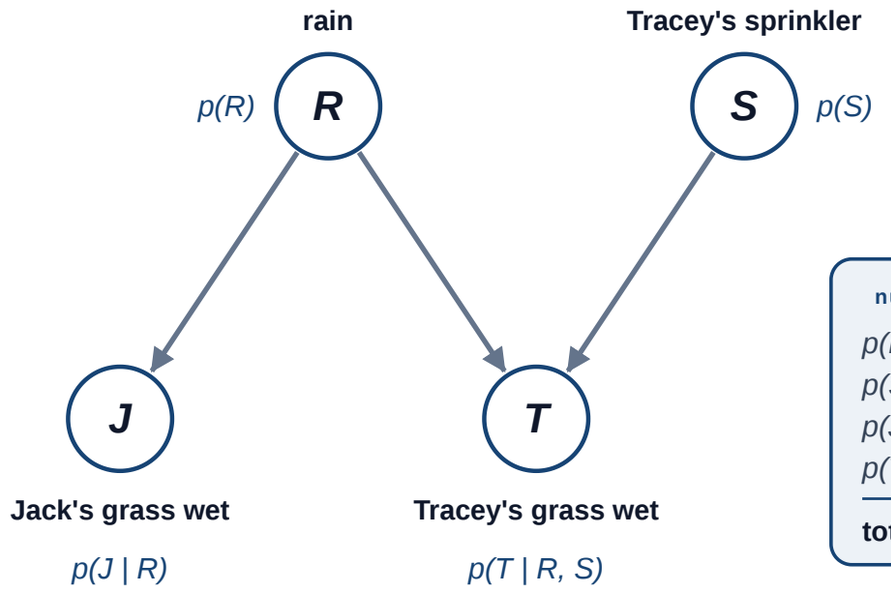
- → Tracey's lawn ignores Jack's, given R and S: $p(T \mid J, R, S) = p(T \mid R, S)$
- → Jack's lawn ignores Tracey's sprinkler: $p(J \mid R, S) = p(J \mid R)$
- → the sprinkler does not change the weather: $p(R \mid S) = p(R)$

> $$p(T, J, R, S) = p(T \mid R, S) \cdot p(J \mid R) \cdot p(R) \cdot p(S)$$

Speaker notes

The reduction is not a trick: it is domain knowledge written as conditional independence. Each assumption can be challenged — ask for one that might fail (a shared hose? then J depends on S). This is "assume it apart" done selectively: three claims instead of naive Bayes' blanket.

The Graph Is the Factorisation



numbers needed	
p(R)	**1**
p(S)	**1**
p(J | R)	**2**
p(T | R, S)	**4**
total	**8**

full joint: 15

factor	$= 1$
$p(R = 1)$	0.2
$p(S = 1)$	0.1
$p(J = 1 \mid R = 0)$	0.2
$p(J = 1 \mid R = 1)$	1.0
$p(T = 1 \mid R = 0, S = 0)$	0.0
$p(T = 1 \mid R = 0, S = 1)$	0.9
$p(T = 1 \mid R = 1, S = 0)$	1.0
$p(T = 1 \mid R = 1, S = 1)$	1.0

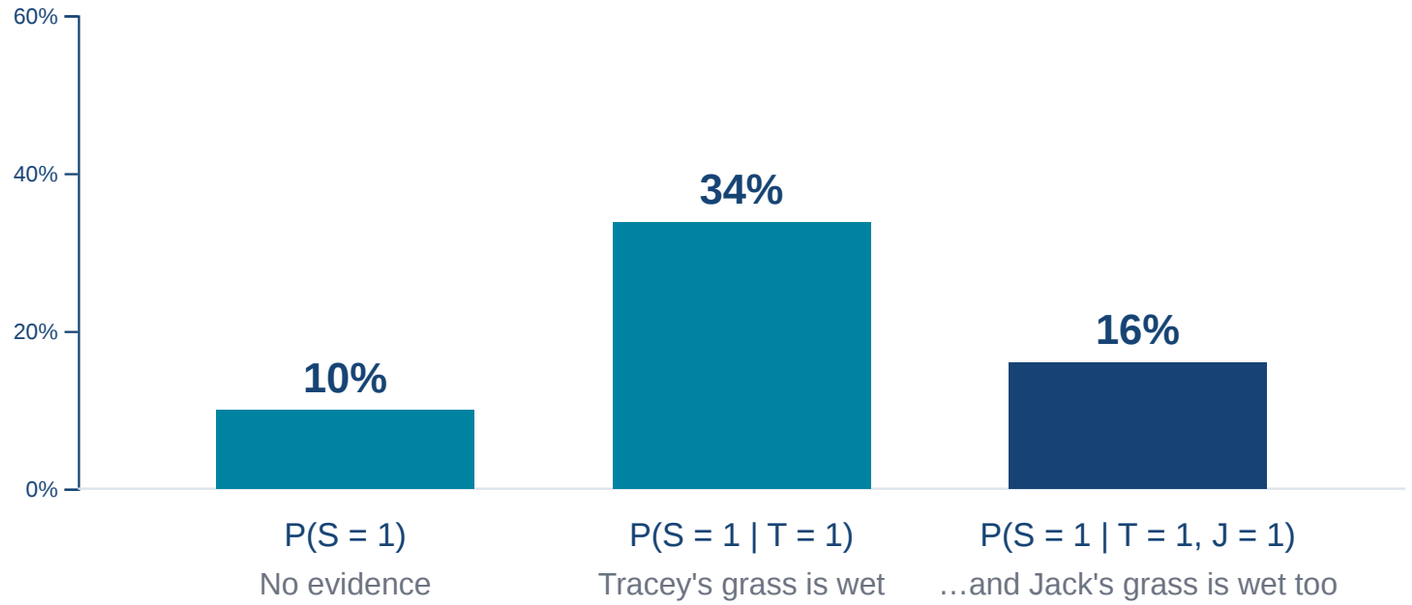
One node per variable, one arrow per conditioning bar, one table per node.

Speaker notes

Read the picture against the formula: every factor p(node | parents) is one node and its incoming arrows. The eight numbers in the table are the eight in the tally of the figure. Point at modelling choices that are visible and challengeable: "grass does not get wet by itself" (0.0), "rain always wets grass" (1.0). Module 5 will ask where such numbers come from.

What Does the Evidence Say?

Rain explains the wetness — the sprinkler falls back towards its prior



Speaker notes

Stepped. Step 1: Tracey's wet lawn raises P(sprinkler) from 10% to 33.8%. Step 2: Jack's lawn is wet too — rain now explains the wetness, and the sprinkler falls to 16.0%: most of the way back, but still above the prior, because it remains a possible additional cause. Compare with the predictions on the board. Computed by exact enumeration of the factorised joint: marginalisation, nothing new.

Explaining Away

$$p(S=1 \mid T=1, J=1) = \frac{\sum_R p(T{=}1 \mid R, S{=}1)p(J{=}1 \mid R)p(R)p(S{=}1)}{\sum_{R,S} p(T{=}1 \mid R, S)p(J{=}1 \mid R)p(R)p(S)} = \frac{0.0344}{0.2144} \approx 0.16$$

Rain and sprinkler are independent: $R \perp\!\!\!\perp S$. Once their **common effect** T is observed they compete to explain it, and evidence for one lowers the other. Conditioning on a common cause removes a dependence; conditioning on a common effect **creates** one.

Speaker notes

The formula is the second click of the viz written out: the numerator keeps S = 1 (0.02 with rain, 0.0144 without), the denominator adds S = 0 (0.18 with rain, 0 without). Name the phenomenon: explaining away. Together with the 16-cell example this gives both directions: conditioning can remove dependence (common cause) and create it (common effect). Module 3 turns the two into a rule that can be read off any graph.

Mini-Exercise

Tracey's lawn is wet. Then she finds the **sprinkler was on**.

Does rain become more or less likely than it was a moment ago?

Speaker notes

The same collider, read the other way; answerable from the previous slide alone. Numbers for checking: P(R=1, T=1) = 0.2, P(T=1) = 0.272, so P(R=1 | T=1) = 0.735; P(R=1, S=1, T=1) = 0.02, P(S=1, T=1) = 0.092.

Mini-Exercise

Tracey's lawn is wet. Then she finds the **sprinkler was on**.

Does rain become more or less likely than it was a moment ago?

Less. The sprinkler explains the wet lawn. $P(R = 1 \mid T = 1) \approx 0.74$ falls to $P(R = 1 \mid T = 1, S = 1) = 0.02/0.092 \approx 0.22$ — almost back to the prior 0.2. Explaining away works in both directions.

Takeaway

Graph the joint: background knowledge becomes conditional independence, and the graph is a picture of how the joint splits — 8 numbers instead of 15. Every *missing* edge is a claim; conditioning can make variables independent (common cause) or dependent (common effect).

What an Arrow Claims

Guiding Question: What exactly does an arrow in a probabilistic model claim — and what does a *missing* arrow claim?

Belief Networks

Bayesian belief network. A **directed acyclic graph** whose nodes are random variables, with one conditional table per node given its parents:

$$p(x_1, \ldots, x_n) = \prod_{i=1}^{n} p\big(x_i \mid \mathrm{pa}(x_i)\big)$$

An arrow permits a dependence. A missing arrow claims an independence: each variable is independent of its non-descendants, given its parents (the **local Markov property**).

Which independence does the wet-grass graph claim for Jack's lawn J?

Speaker notes

Math first — the students built one by hand in Module 1, so the general form reads as a summary. "Acyclic" matters: with a cycle the product would not define a distribution.

This answers the module's guiding question. An arrow only *permits*: the table may ignore the parent. In the wet-grass tables, p(T = 1 | R = 1, S) is 1 for both values of S — given rain, the sprinkler arrow carries no influence at all. Absence of an arrow, by contrast, forces an independence for every table one could put on the graph. So the claims of a graph live in its missing edges.

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Which independence does the wet-grass graph claim for Jack's lawn J?

$J \perp\!\!\!\perp \{S, T\} \mid R$: once we know whether it rained, nothing else says more about Jack's lawn.

Counting the Numbers

model, binary variables	full joint	graph
wet grass, 4 nodes	15	$1 + 1 + 2 + 4 = 8$
naive Bayes, class + 30 flags	$2^{31} - 1 \approx 2.1 \cdot 10^9$	$1 + 2 \cdot 30 = 61$
20 nodes, at most 3 parents each	$2^{20} - 1 \approx 10^6$	at most $20 \cdot 2^3 = 160$

Where does the saving come from — and what makes it disappear?

Speaker notes

A table because the three models are compared along the same two columns. Check the naive-Bayes row against Lecture 03: one prior plus one Bernoulli per flag and class. The last row is the general bound: 20 tables of at most 8 numbers each, against 1,048,575.

The second half of the answer prepares the next slide: many parents is the one place where a graph does not help by itself.

Counting the Numbers

model, binary variables	full joint	graph
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Where does the saving come from — and what makes it disappear?

A node with k binary parents needs 2^k numbers. The saving comes from **missing arrows**; it disappears as soon as some node has many parents.

Keeping Tables Small: Noisy-OR

A symptom y with many possible causes $x_1, \ldots, x_k$: **each present cause fires on its own** with probability λ_i, and y occurs if any fires:

$$p(y = 0 \mid x_1, \ldots, x_k) = \prod_{i : x_i = 1} (1 - \lambda_i)$$

k numbers instead of 2^k. Two causes present, $\lambda = 0.8$ and 0.6: $p(y = 1) = 1 - 0.2 \cdot 0.4 = 0.92$.

THE ASSUMPTION BOUGHT

Causes act **independently** — no interaction. Two drugs that are harmless alone but dangerous together break it.

Divorcing is the other trick: insert a summary node ("total load") between a group of parents and the child.

Speaker notes

Every table-shrinking technique is another assumption, and the slide names it. Ask for a real case of interaction before showing the warning. Mention that noisy-OR as written has no "leak": with no cause present, y never occurs; a leak term λ_0 adds the unexplained cases.

Divorcing is an elicitation argument as much as a storage one: an expert can state four numbers, not sixty-four.

Mini-Exercise

Tracey and Jack share a garden hose: add the arrow $S \to J$ to the wet-grass network.

How many numbers does the network need now — and which independence claim is gone?

Speaker notes

One arrow, one table, one claim. Head-computable: only J's table changes. The second half is the module's thesis in miniature: adding an arrow removes a claim and costs numbers; removing one makes a claim and saves them.

Mini-Exercise

Tracey and Jack share a garden hose: add the arrow $S \to J$ to the wet-grass network.

How many numbers does the network need now — and which independence claim is gone?

10. $p(J \mid R, S)$ needs 4 numbers instead of 2: $1 + 1 + 4 + 4$. Gone is $J \perp\!\!\!\perp S \mid R$ — the missing arrow was that claim.

Takeaway

A belief network is **a DAG plus one table per node**. An arrow only permits a dependence; every **missing** arrow claims an independence, and each claim saves numbers. Shrinking tables further — noisy-OR, divorcing — buys more savings with more assumptions, which should be said out loud.

Reading Independence off the Graph

Guiding Question: Two causes are independent until you observe their shared effect. Can we see such things in the picture — before computing a single number?

Three Ways Three Nodes Connect

CHAIN

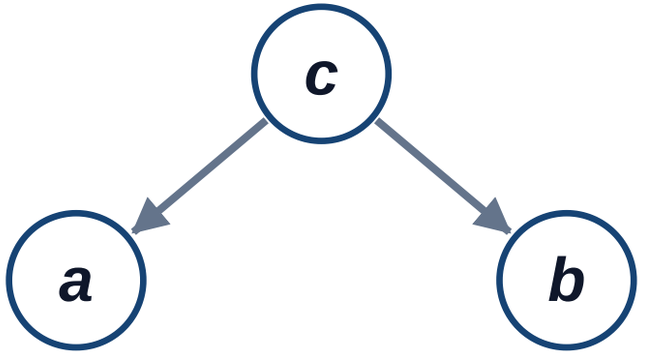


a, b: **dependent**

given c: **independent**

observing c blocks the path

FORK · COMMON CAUSE

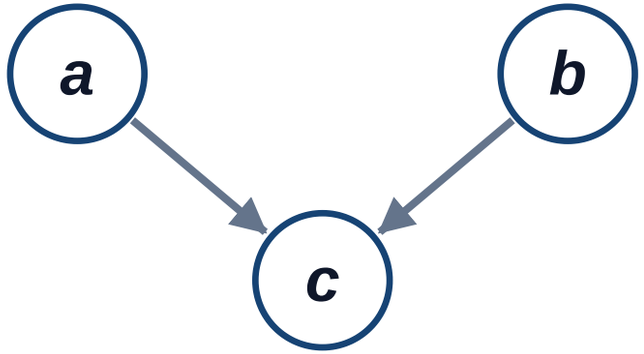


a, b: **dependent**

given c: **independent**

observing c blocks the path

COLLIDER · COMMON EFFECT



a, b: **independent**

given c: **dependent**

observing c opens the path

Every independence question in a DAG reduces to these three.

Speaker notes

Two of the three were met in Module 1: the fork is the common cause (temperature → ice cream, drownings; the 16 cells), the collider is the common effect (rain → wet lawn ← sprinkler). The chain is new but intuitive: information flows along it until the middle is known. Emphasise the collider's inversion: conditioning normally blocks, here it opens.

d-Separation

A path between X and Y is **blocked** by a set Z if it contains

- a chain or fork whose middle node **is in** Z, or
- a collider whose middle node — and every descendant of it — is **not in** Z.

If every path is blocked, X and Y are **d-separated** by Z, and then $X \perp\!\!\!\perp Y \mid Z$ in every distribution that factorises over the graph.

Sound, not complete: a particular table may add independences the graph does not show — in our tables, once it rains, Tracey's lawn ignores the sprinkler.

Speaker notes

The definition after the three cases, so it reads as a summary. "Every descendant": observing a consequence of the collider is partial evidence about the collider itself, so it opens the path too — the mini-exercise tests it. The note ties back to "an arrow permits": the graph guarantees the independences it shows, and a table can make more hold numerically.

d-Separation on the Wet Grass

The only path from J to S: $J \leftarrow R \rightarrow T \leftarrow S$ — a fork at R, a collider at T.

query	path	verdict	the numbers
$J \perp\!\!\!\perp S$?	collider T unobserved: blocked	independent	$P(J{=}1) = 0.36 = P(J{=}1 \mid S{=}1)$
$J \perp\!\!\!\perp S \mid T$?	collider T observed: open	dependent	$P(J{=}1 \mid T{=}1) = 0.79$, but 0.37 if also $S{=}1$
$J \perp\!\!\!\perp S \mid T, R$?	fork R observed: blocked	independent	$P(S{=}1 \mid T{=}1, R{=}1) = 0.1$, with or without $J{=}1$

> The middle row is Module 1's explaining away: $10\% \rightarrow 34\% \rightarrow 16\%$ for the sprinkler. The graph predicted it before any number was computed.

Speaker notes

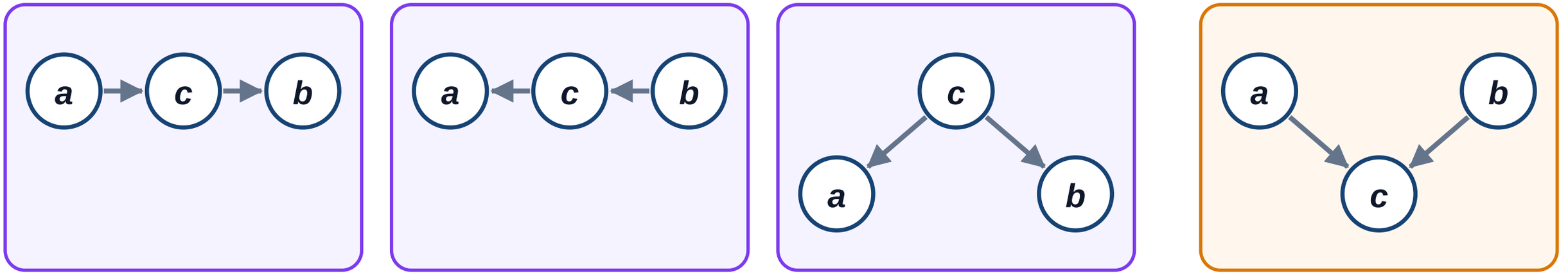
Walk the path once, triple by triple, for each query. The numbers are exact (enumeration of the factorised joint): P(J=1) = 0.2 · 1 + 0.8 · 0.2 = 0.36; P(J=1 | T=1) = 0.2144/0.272 = 0.788; P(J=1 | T=1, S=1) = 0.0344/0.092 = 0.374; given rain, the sprinkler is back at its prior 0.1 whatever Jack's lawn does.

The point to stabilise: d-separation answers "does this evidence matter for that query?" from the picture alone — which is also what an inference algorithm uses to ignore irrelevant parts of a network.

Arrows Are Not Causes

same claim: a, b independent given c

different claim



data cannot tell these three apart

the direction of the arrows is not in the data

a, b independent;
dependent given c

Same skeleton, same colliders: **the same independences** (Markov equivalence). Data cannot orient such arrows — they are **not** causal claims.

Speaker notes

This is the honest answer to the guiding question of Module 2. The arrows of the wet-grass graph were drawn in the causal direction because we *know* the causal story; the independences would allow other directions. Only a collider leaves a trace in the data (two independent variables that become dependent given a third), which is why structure-learning algorithms can orient some edges and not others. Causal claims need more than a fitted graph — interventions or assumptions stated outside the data. Module 5 returns to this under structure learning.

Mini-Exercise

Tracey's doormat gets wet whenever her lawn is: add $T \rightarrow D$. You see only a **wet doormat —** not the lawn.

Are rain and sprinkler still independent?

Speaker notes

One question, one rule — the "descendant" clause of d-separation, which students tend to skip. If the room is quick: what if we also learn it rained? Then the sprinkler becomes less likely, exactly as on the lawn.

Mini-Exercise

Tracey's doormat gets wet whenever her lawn is: add $T \to D$. You see only a **wet doormat —** not the lawn.

Are rain and sprinkler still independent?

No. D is a descendant of the collider T, so observing it opens the path $R \to T \leftarrow S$: a wet doormat also makes rain and sprinkler compete to explain it.

Takeaway

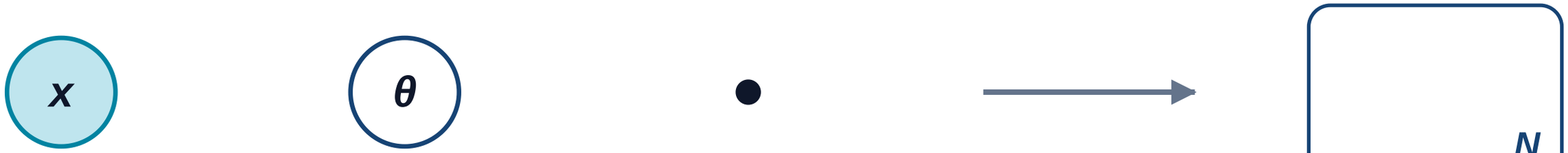
Independence in a belief network is a **property of the graph**: chains and forks are blocked by observing the middle, colliders are opened by it (or by a descendant). d-separation applies that to every path. What it cannot read off is the direction of cause: equivalent graphs make the same claims.

Drawing Repeated Structure

Guiding Question: A model of 10,000 observations should not need 10,000 nodes. How is a repeated structure drawn — and what does the drawing claim about the observations?

Drawing a Model

A probabilistic model is a factorisation, and the diagram is its picture. Five conventions carry almost all of it:



shaded
observed: in the data

open
latent or parameter

solid dot
fixed, not inferred

arrow
child conditioned on parent

plate
repeat N times

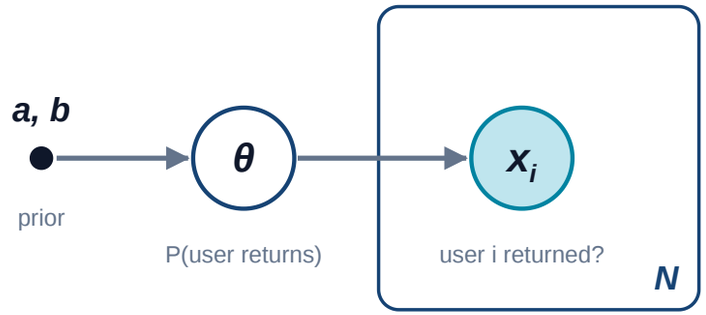
> Nothing new is claimed: the arrows are read exactly as in a belief network. The shading only says **what the data fixes** — which variables we condition on.

Speaker notes

Insist that this is notation, not theory: every diagram in a paper is a factorisation students can already read. Shading is the convention they will need most: conditioning on the data, Lecture 01's second move, is literally "shade the observed nodes".

Plates

A **plate** is a box with a count in its corner: *repeat what is inside N times*. A model of 10,000 users does not need 10,000 nodes.
The copies are **independent given what sits outside** — the parameters, which learning estimates. Nested plates draw grouped data: K hospitals with N_k patients each.

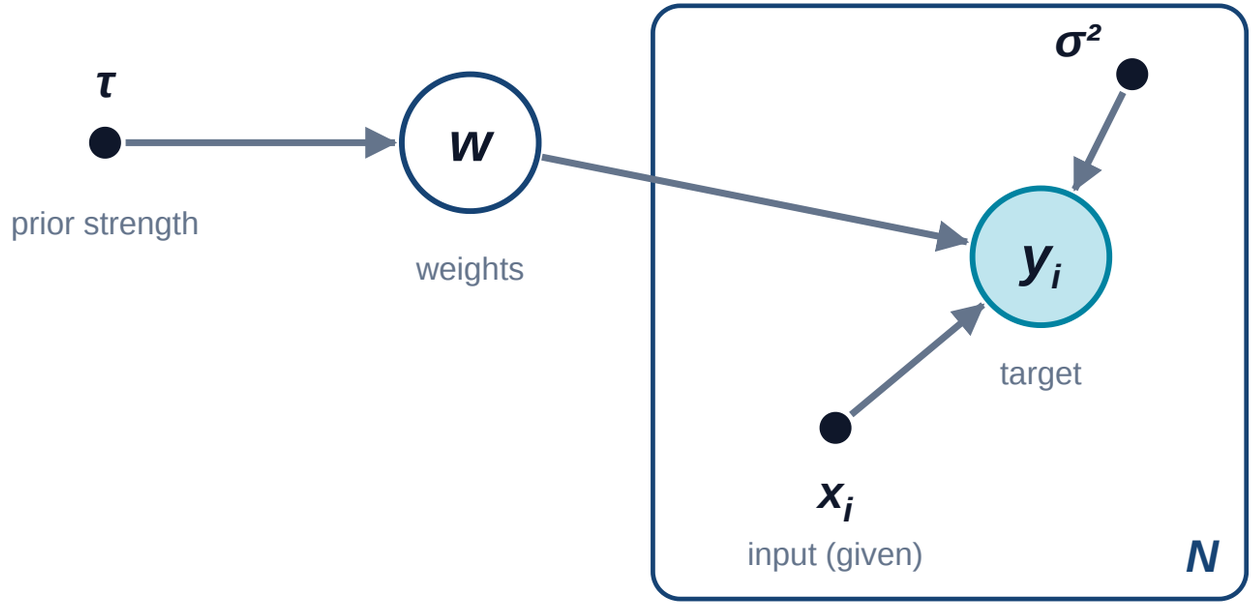


> Given θ the users are independent. With θ unknown they are **not**: in Lecture 02, the first ten users changed our prediction for the eleventh. That dependence is exactly why learning from data works.

Speaker notes

The i.i.d. assumption of Lecture 02 is *conditional* i.i.d.: independent given the parameters. The figure is the L02 model as a plate: θ (open) → x_i (shaded) inside a plate of N, with a dot for (a, b) outside. Marginally, after summing θ out, the x_i are dependent — the Beta–Bernoulli predictive of the next user depends on how many earlier users came back. A student who says "i.i.d. means the data cannot teach us anything about the next point" has missed the word "given".

Reading a Diagram



Read off the factorisation:

$$p(w \mid \tau) \prod_{i=1}^{N} p(y_i \mid x_i, w, \sigma^2)$$

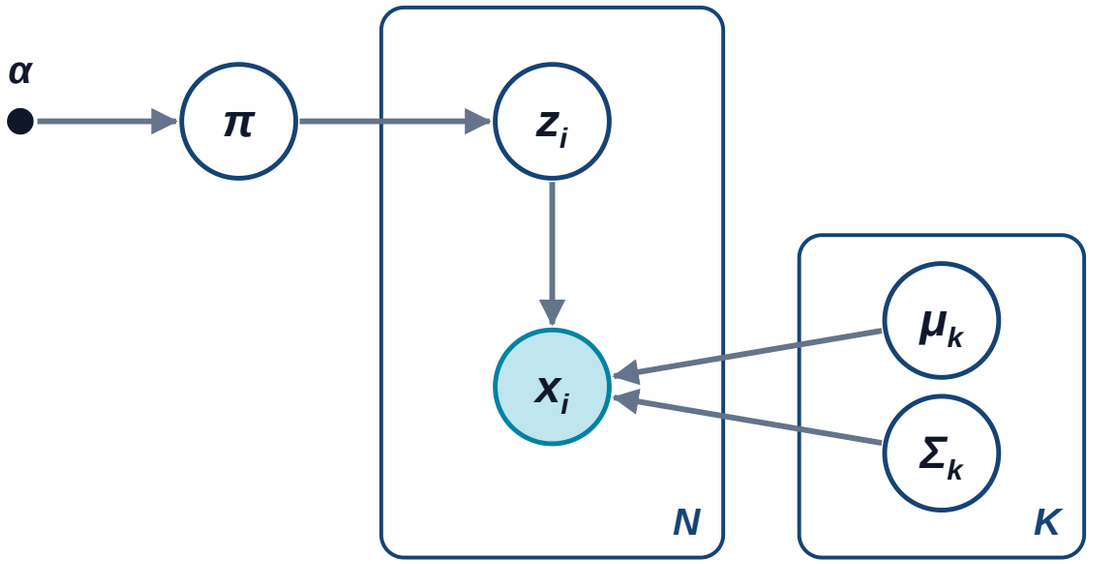
The plate became the product; each arrow became a conditioning bar.

x_i is a **dot**: the inputs are given, not modelled — this is a discriminative model, $p(y \mid x)$, with a prior on w. Lecture 07 conditions on the shaded y_i and keeps the whole posterior over w.

Speaker notes

The Bayesian linear regression of Lecture 07, drawn now so that the model is familiar before its inference is. Two details the drawing forces us to decide: the inputs are fixed (a dot, no factor p(x_i) — Lecture 04's "only p(y | x)" visible in the notation), and the noise variance σ^2 is fixed rather than learned. Ask what would change if σ^2 were an open node: it would get a prior and a posterior like w.

Mini-Exercise

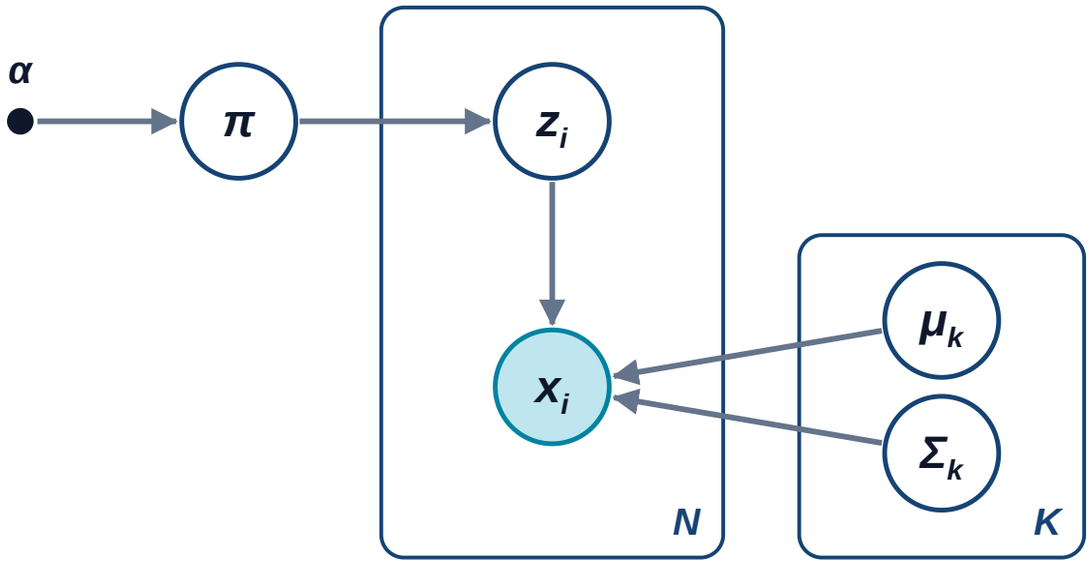


Which nodes does learning have to infer — and why is α not among them?

Speaker notes

One question, read off the conventions slide. The factorisation in the solution is for checking and for the notes: every open node with no parent inside a plate gets a prior (π, μ_k, Σ_k), every arrow a conditioning bar, and x_i depends on *all* components through z_i, which selects one. The z_i are latent *per data point* — the hidden labels that EM will infer in Lecture 07.

Mini-Exercise



Which nodes does learning have to infer — and why is α not among them?

π, every μ_k, Σ_k and every label z_i — all **open**.

α is a **dot**: fixed by us. Only the x_i are observed:

$$p(\pi \mid \alpha) \prod_k p(\mu_k, \Sigma_k) \prod_i p(z_i \mid \pi) p(x_i \mid z_i, \mu, \Sigma)$$

A **mixture of Gaussians** — Lecture 07.

Takeaway

A model diagram is a factorisation you can see: **shading** marks what the data fixes, **plates** mark what repeats independently *given* what sits outside, and **what sits outside** is what learning has to estimate.

Can We Compute With It?

Guiding Question: We can now write the joint down. Can we actually condition on it — and where do its numbers come from?

Three Questions You Can Ask a Model

question	on the wet grass	with complete data
infer unobserved variables	$P(S \mid T, J)$	cost set by the graph's shape
learn the parameters	the 8 table entries	counting
learn the structure	which arrows at all?	hardest — and only up to equivalence

Speaker notes

The three tasks, one row each, with the running example. The ordering of difficulty matters and is often stated wrongly: with complete data, parameter learning is the easiest (it is counting); inference can be hard; structure learning is the hardest. The module takes them in the order inference → parameters → structure.

Inference by Elimination

Explaining away needed $P(T{=}1, J{=}1)$. Sum the factorised joint — and **push each sum as far right as it goes**:

$$P(T{=}1, J{=}1) = \sum_R p(R)p(J{=}1 \mid R) \underbrace{\sum_S p(S)p(T{=}1 \mid R, S)}_{m(R)}$$

Inner sum first

$m(1) = 0.1 \cdot 1 + 0.9 \cdot 1 = 1$

$m(0) = 0.1 \cdot 0.9 + 0.9 \cdot 0 = 0.09$

Then the outer one

$0.2 \cdot 1 \cdot 1 + 0.8 \cdot 0.2 \cdot 0.09$

$= 0.2 + 0.0144 = 0.2144$ ✓

Speaker notes

Math first: show the formula, ask why it equals the plain double sum (distributivity: factors that do not mention S can leave the inner sum). The result is the denominator of the explaining-away formula in Module 1 — the same number, computed smarter. m(R) is a small table, a "message" from S to R; this is variable elimination.

On four variables the saving is invisible. The next slide shows where it becomes everything.

Why Pushing Sums In Pays

A chain of n binary variables $X_1 \to X_2 \to \cdots \to X_n$. To get $p(x_n)$:

$$2^{n-1}$$

terms if every other variable is summed at once — for $n = 30$, about 537 million

$$4(n-1)$$

operations if the sums are pushed in, one variable at a time — for $n = 30$, 116

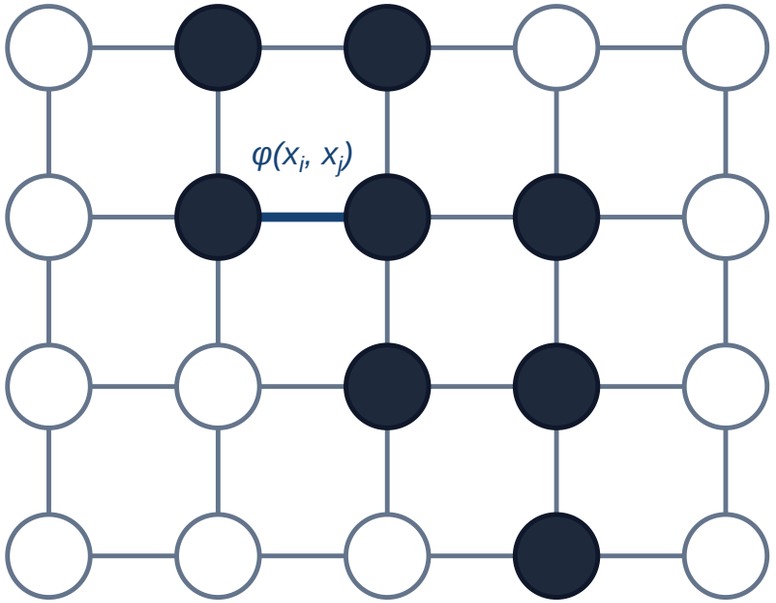
Eliminating X_1, then X_2, … passes a **two-entry table** along the chain. The cost is set by the **largest table** created on the way — by the graph's shape, not its size.

Speaker notes

The chain is the best case: each elimination step multiplies a two-entry message into a 2 × 2 table and sums one variable out — four products. Trees behave like chains. Numbers: 2^29 = 536,870,912 against 4 · 29 = 116.

The general statement (not for the exam): the cost is exponential in the treewidth of the graph, roughly the size of the largest table elimination has to build. The next two slides meet a graph where that table is enormous.

Undirected Models



each edge: a potential that prefers neighbours to agree

Pixels influence each other **symmetrically**. A **Markov network** puts a non-negative **potential** ϕ on each edge and normalises once, globally:

$$p(x) = \frac{1}{Z} \prod_{(i,j)} \phi(x_i, x_j), \qquad Z = \sum_x \prod_{(i,j)} \phi(x_i, x_j)$$

Two pixels, $\phi = 3$ if equal, 1 if not: $Z = 3 + 1 + 1 + 3 = 8$, so $P(\text{equal}) = 6/8$.

Speaker notes

Only what the lecture needs: potentials are not probabilities and do not normalise locally, so a global Z is required. The two-pixel example makes Z concrete and head-computable. For an image, Z sums over every possible image: 2 to the number of pixels. That sum is the price of modelling symmetric influence — and the subject of the next-but-one slide.

Directed and Undirected

	belief network	Markov network
edges	directed, acyclic	undirected
factors	conditional tables, normalised locally	potentials, one global Z
natural for	causal or generative stories	symmetric interaction: pixels, neighbours
observing a common effect	creates dependence	— (no colliders)

Inference runs on the undirected form: **moralisation** joins parents that share a child (R – S in the wet grass) and drops the arrow heads; elimination works on that graph.

Speaker notes

Two slides for Markov networks, by design (course plan default): what they are, and how they relate to belief networks. The relation that matters for this module is the compilation one: exact inference algorithms (elimination, junction trees) turn a belief network into its moral graph first. The moral graph of the wet grass joins R and S — it has forgotten that they are marginally independent, which is why independence is read on the DAG and computation done on the moral graph. Neither model class contains the other: a collider has no undirected equivalent, a four-cycle no directed one.

Writable Is Not Queryable

A 100×100 black-and-white image as a pixel lattice:

79,200

numbers to **write it down**: 19,800 edges, 4 per potential

$$2^{100} \approx 10^{30}$$

entries in the table that elimination must build — a whole row of pixels at once

Exact inference in belief networks is **NP-hard** in general, and so is approximate inference with a guaranteed error bound. What remains: exploit the shape where it is kind (chains, trees), and otherwise **sample** or **approximate** — trading the guarantee for an answer.

Speaker notes

This is the lecture's thesis, shown rather than asserted. The model is tiny to state — one 2 × 2 potential per neighbouring pair — and hopeless to condition exactly: eliminating the pixels row by row leaves a table over an entire row, 2^100 entries (the treewidth of an n × n grid is n). Z itself sums over 2^10,000 images.

Hardness results: exact inference, Cooper (1990); approximate inference to a guaranteed accuracy, Dagum & Luby (1993). In practice image models use sampling (MCMC) or variational approximations; Lecture 07's EM is a first, exact-where-possible example of alternating between inference and fitting.

Learning the Parameters Is Counting

Fifty observed days. Estimate Jack's table $p(J{=}1 \mid R)$:

	days	Jack's lawn wet	MLE (peak)	$+1$ pseudo-counts (mean)
$R = 0$	40	8	$8/40 = 0.20$	$9/42 \approx 0.21$
$R = 1$	10	10	$10/10 = 1.0$	$11/12 \approx 0.92$

> Counting **is** maximum likelihood (L03); pseudo-counts **are** the Beta posterior mean (L02) — Laplace smoothing (L03). "Rain always wets the grass" was an MLE on ten days.

Hidden variables leave no counts: infer them, re-count, repeat — EM, Lecture 07.

Speaker notes

The course's spiral on one slide. With complete data the likelihood of a belief network splits into one categorical per node and parent configuration, so each row is fitted separately — the same argument as naive Bayes, which is the star-graph special case. The 1.0 in Module 1's table now looks different: it is the three-heads-in-three-flips problem of Lecture 03 in a table row. A parent configuration that never occurs leaves its row undetermined (0/0); the pseudo-counts fill it with the prior mean 1/2.

Learning the Structure

Score and search

Try graphs, score each by how well it fits the data *per parameter*, keep the best. The number of DAGs grows super-exponentially.

Test independences

Run conditional-independence tests and keep the graphs consistent with them — d-separation, read backwards.

Either way, data determines a graph only **up to Markov equivalence**: arrows between equivalent graphs are not in the data. A learned structure is not a causal discovery.

Speaker notes

One slide on purpose: structure learning is a research area, not a technique to apply on Monday. The warning is Module 3's "arrows are not causes", now as a limit of learning. Students will meet confident causal claims from fitted graphs; this is the sentence that lets them challenge one.

Mini-Exercise

Jack's lawn was wet on all **4** rainy days in a small sample.

What does counting give for $p(J{=}1 \mid R{=}1)$ — and what does one pseudo-count on each side give?

Speaker notes

Head-computable, and the same lesson in three lectures: L02 (Beta posterior mean), L03 (the peak can lie; smoothing). If time: with 400 of 400 the two readings agree to three decimals — the prior washes out.

Mini-Exercise

Jack's lawn was wet on all **4** rainy days in a small sample.

What does counting give for $p(J{=}1 \mid R{=}1)$ — and what does one pseudo-count on each side give?

Counting: $4/4 = 1$ — "rain always wets the lawn", from four days. **Smoothed:** $5/6 \approx 0.83$, the posterior mean under $\mathrm{Beta}(1,1)$. Same data, two readings: the peak and the mean.

Takeaway

A graphical model makes the joint **writable**; its **shape** decides whether it is **queryable** — elimination is cheap on chains and trees and hopeless on a lattice. With complete data the parameters are counts: the MLE, or with pseudo-counts the Beta posterior mean. The structure is learnable only up to equivalence.

Appendix

Uncertain and Unreliable Evidence

Uncertain (soft) evidence

We are not sure of the observation itself: a distribution $\tilde{x}$ over its states. Sum over them:

$$p(y \mid \tilde{x}) = \sum_{x} p(y \mid x) p(x \mid \tilde{x})$$

We look at the lawn from the window without touching it: $p(T{=}1 \mid \tilde{T}) = 0.8$.

Unreliable evidence

We keep the report but **replace** its table by our own judgement of how reliable it is:

$$p(x \mid \mathrm{pa}(x)) \to \tilde{p}(x \mid \mathrm{pa}(x))$$

Jack says his lawn is wet, but he did not check.

Both are drawn with a **dashed** edge.

Speaker notes

Appendix (course plan default): niche for this audience, kept for questions. The distinction: uncertain evidence keeps the observation and softens it; unreliable evidence keeps the report but changes the model of how the report arises. Students who conflate the two model rumours as noise.